PAUL-PETER ARSLAN

Paris, France | paulpeterarslan@gmail.com | polpii97@mit.edu | +33 7 83 59 65 83

Research Scientist with a PhD focus on large language model systems, multi-agent architectures, and rigorous behavioral evaluation of human-AI interaction. My work is driven by the conviction that the next generation of AI should be built to genuinely understand and serve diverse communities, through responsible engineering and breakthrough research. I build end-to-end LLM systems, from multi-agent reasoning pipelines to behavioral evaluation, with the goal of creating AI that advances science and improves how people learn, create, and recover.

RESEARCH EXPERIENCE

Visiting Research Student, MIT Media Lab (Tangible Media Group) **Cambridge, MA, USA**

• Co-first author (equal contribution), Tangible Co-Ideation: multi-agent LLM system with multimodal RAG, fine-tuning and prompt engineering. Within-subjects user study (N=32, 16 pairs) compared tangible vs. screen-only conditions with statistically significant effects on verbal interaction and ideation behaviour. Advised by Prof. Hiroshi Ishii. Submitted to ACM UIST 2026; ACM DIS 2026 SDC finalist.
• Coursework: Interaction Intelligence: Teleabsence (Prof. H. Ishii), Design Intelligence (Prof. M. Coelho)
• Winner, Connect track at MIT Hard Mode: Hardware AI Hackathon 2026 (IPheromone: wearable multi-agent LLM system for proximity-based human compatibility, agent-to-agent reasoning)

Ph.D. Researcher, Institute for Future Technologies / Neuroscience Institute Paris **Paris, France**

Supervisors: Dr. Xiao Xiao (Institute for Future Technologies / MIT), Dr. Pavel Lindberg (INSERM) 2023 – Present

• Tangible Co-Ideation: multi-agent LLM system (co-first author, equal contribution): AI Mentors as LLM-simulated expert perspectives on text and image, grounded by a multimodal RAG knowledge base (OCR + cross-modal retrieval); fine-tuning and prompt engineering align each agent with its persona voice. Related-work module queries arXiv, Semantic Scholar, and CrossRef. Evaluation: within-subjects user study N=32 (16 pairs), tangible vs. screen-only, on verbal interaction.
• IPheromone: personalized LLM agents (team project): Wearable Raspberry Pi system; each user has a personal LLM agent that builds a structured profile from conversation; agents exchange profiles and reason about compatibility, producing match scores. End-to-end functioning prototype on real hardware. Winner, Connect track at MIT Hard Mode Hardware AI Hackathon 2026.
• HD-EMG Deep Learning: high-density (224-channel) signal decoding with a causal Transformer (FITLight, FFT-based attention) and a GRU branch (MFTEventGRU); full PyTorch pipeline (dataset, training loop, mixed-precision, hyperparameter sweeps, with low-latency inference suitable for biofeedback.
• ReTouche: Projection-AR piano-learning system; mixed-methods evaluation: 3-arm comparative study (n=18) + longitudinal autoethnography + expert focus group. ACM CHI 2026
• Rhythm Karaoke: Millisecond-resolution timing engine for fine motor behavior; powers downstream ML feature extraction. Submitted to Springer Nature (Scientific Reports)
• 2nd-author contributions: TMS-based motor mapping, haptic feedback prototypes, embedded firmware, and front-end interfaces for collaborator studies
• Teaching: Web Development, AI with Bio-sensors, Research Workshops for Master students

PUBLICATIONS

First Author

Peer-Reviewed

• **Arslan, P.-P.**, Xiao, X., et al. ReTouche: Embodied Representations for Self-Directed Piano Learning. *ACM CHI 2026* (full paper, accepted).

Under Review

• Li, Y. L.*, Kuang, Q.*, **Arslan, P.-P.***, Xiao, X., Ishii, H. Tangible Prompting: A Physical Vocabulary for Navigating Conceptual Spaces in Human-AI Co-Ideation. Submitted to *ACM UIST 2026*; Demo accepted at *ACM CHI 2026*. (*equal contribution)
• Li, Y. L.*, Kuang, Q.*, **Arslan, P.-P.***, Xiao, X., Ishii, H. Tangible Co-Ideation: Designing Embodied Prompting for Creative Thinking with Large Language Models. *ACM DIS 2026 SDC* (finalist). (*equal contribution)
• **Arslan, P.-P.**, et al. A Novel Method for Rhythmic Imitation of Finger Movements and the Effects of Auditory Stimuli and Feedback. Submitted to *Scientific Reports* (Springer Nature).

In Preparation

• **Arslan, P.-P.**, et al. Predicting Continuous Finger Forces from High-Density EMG with AI Regression Models. In preparation (target: *Nature*-family journal).
• **Arslan, P.-P.**, et al. Rhythmic Finger Tapping Performance in Stroke Survivors: Effects of Auditory Stimuli and Feedback. In preparation.

Co-Authored

Peer-Reviewed

• Dziezuk, E., **Arslan, P.-P.**, et al. (2025). Mécanismes sensoriels de la récupération de la dextérité manuelle après AVC : protocole d'une étude de cohorte prospective. *Kinésithérapie Scientifique*.

Under Review

• Dziezuk, E., Badr, L., **Arslan, P.-P.**, et al. A novel haptic stimulation device to improve finger movements – a validation and reliability study in healthy subjects. Submitted to *IEEE Transactions on Haptics* (Feb. 2026).

In Preparation

• Duan, Z., **Arslan, P.-P.**, Plantin, J., Lindberg, P. G., Wang, R. Motor Unit Coherence and Discharge Behavior During Single-Finger Isometric Contractions After Stroke. In preparation.

EDUCATION

Master Degree in Innovation, Research & Manufacturing *Paris, FR*
IFT ESILV Leonard da Vinci Engineering School *2021*
Relevant coursework
Advanced algorithms, Big Data Infrastructure, Network Architecture, Deep Machine Learning, Computer Graphics, Advanced electonic and mechanic, Soft and Active Matter
UQAC – *international exchange* Canada
Web development, statistic computing, marketing *Sept 2018 – Dec 2018*
GPA: 3.96 / 4

SKILLS

Machine Learning & AI: PyTorch · TensorFlow / Keras · scikit-learn · Transformer architectures · CNN, GRU, LSTM · Large Language Models · Retrieval-Augmented Generation (RAG) · Prompt engineering & multi-agent orchestration · Signal processing & neural decoding · Real-time inference pipelines

Software Engineering: Python · TypeScript / JavaScript (React, Next.js, Node.js) · C# · MATLAB, R · REST APIs · Git, Docker, Linux · Unity · Arduino, Raspberry Pi, embedded firmware · CAD & rapid prototyping

Languages: French (native), English (fluent, academic writing & presentations), Spanish (conversational)

WORK EXPERIENCE

DEVO, Robotic startup Paris, FR
Creation and development of a company *September 2020 – Aug 2021*
- Autonomous mobile robot visual SLAM based
- Development of a front-end interface and a back-end to manage a robot swarm
- Make a website and a video for commercial interests

Learning Planet Institute, Interdisciplinary Research Center Paris, FR
Research Engineer *Mar 2020 – Aug 2020*
- In charge of the development of a mobile application
- Development of a pedagogical concept to teach math to children through games
- *Game development with Unity and application on a connected gymnasium*

Usine.io - STATION F Paris, FR
Electronic Engineer *Jun 2019 – Jul 2019*
- Coaching of 8 teams of 6 students in the IMT Artificial Intelligence Challenge
- Hardware project development/ fast prototyping with 3D printer, laser cut, electronic components, 3D modeling software
- Design thinking with project initiators analyzing market application

Instant Gaming Barcelona, SP
Developer and project manager *Jun 2018 – Aug 2018*
- Create a new swimming pool alarm system (Raspberry, Electronic, Machine learning)
- Project management
- Communicated efficiently with management and staff

Sue Ryder Oxford, UK
Volunteer, sales assistant *Jul 2018*
- Worked register (Cash management, customer interaction)
- Collect and prepare the donations for the shop

ACTIVITIES

LCE Chess club La Ciotat, FR
Chess instructor *2007 – 2018*
- DIFF certificate (French Chess Federation)
- FIDE Elo rating: 1880 (World Chess Federation)
- Instructed a senior class of 10 students
- 2nd place, PACA Open; 1st place, National IV Team Competition

Eagles, ESILV US Football Team **Paris, FR**
Wide Receiver *2017 – 2018*
- University French Champion (2017), US Football Association

Sports: Rugby, Skiing, Swimming, Krav Maga

Music: composition with FL Studio (Fruity Loops)